

Lab 4

Jonathan Tadesse.

Collaborated with Yifan Feng and Alexander Konstantinov

1

We simulated the interference with different N mesh points $N \in (2^{13}, 2^{14})$ and different time steps dt . The figures down below with $N = 2^{13}$ and $N = 2^{14}$ mesh points respectively seem to be very similar. Furthermore, both figures below do not deviate that much from the original plot. This made us conclude that the results are real and not dependent on numerical parameters.

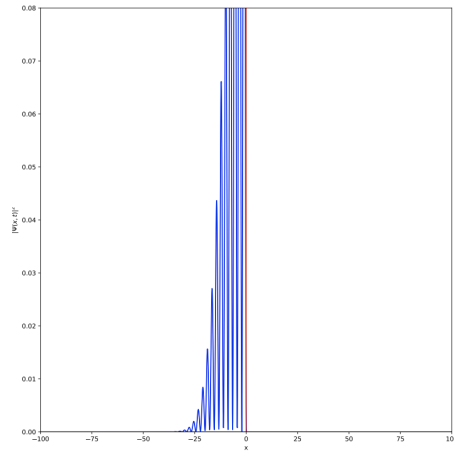


Figure 1: Interference pattern for $N = 2^{13}$

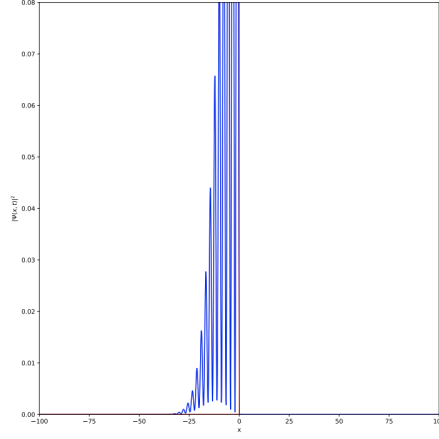


Figure 2: Interference pattern for $N = 2^{14}$

2

We simulated the wave packet's interaction with the new potential $V(x) = 0$ for $x < 1$ and $V(x) = \frac{1}{x}$ for $x \geq 1$ (Coulomb barrier). Figure 3 shows the resulting probability density, showing the splitting of the wave packets density into two, one part of the wave packet is reflected, while the other part successfully tunnels through the barrier. .

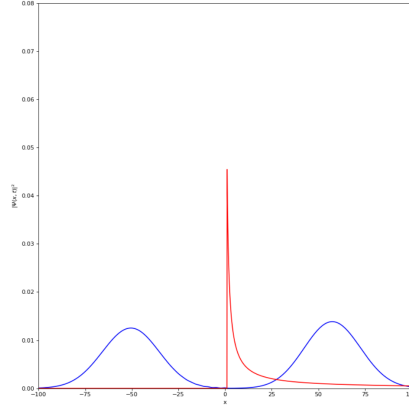


Figure 3:

To calculate the transmission probability T , we use the following relation:

$$T = \frac{Y_T}{Y_R + Y_T}.$$

Where Y_R is the maximum amplitudes when $x < 1$ and Y_T when $x > 1$.

According to Gamow's theory, T should vary exponentially with energy E :

$$T \propto \exp\left(-\frac{k}{\sqrt{E}}\right)$$

We plotted $\ln T$ against $1/\sqrt{E}$. The result is shown in Figure 3 ,we expected a linear plot.

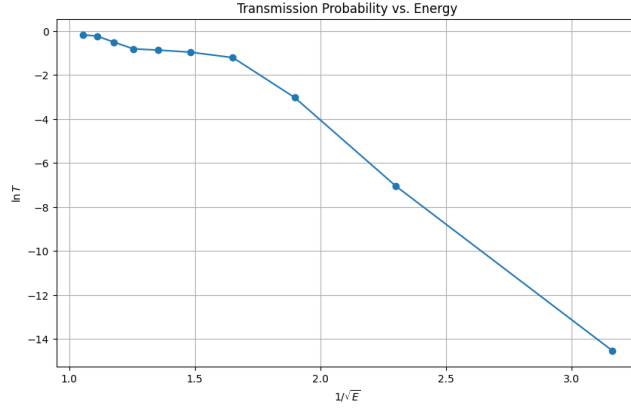


Figure 4: Transmission probability vs Energy

The plot seem to show an almost linear graph for $1/\sqrt{E} \in (1.5, 3)$ agreeing with the Gamow's theoretical prediction. Outside $1/\sqrt{E} \in (1.5, 3)$ we see deviation from the linear path. .

3

We initialized a gaussian wave-packet with energy $E = 0.9$. The potential $V(x)$ was defined such that $V(x) = 0$ for $x < 0$ and $V(x) = 1$ for $x \geq 0$. We simulated the evolution of the wavepacket and figure 5 is the result. As we can see the wave packet encounters the potential step at $x = 0$. Since the energy $E = 0.9$ is less than the barrier height $V_0 = 1$, the wave packet is reflected to the right. The probability density on the right side of the potential step is zero, which means there is no transmission.

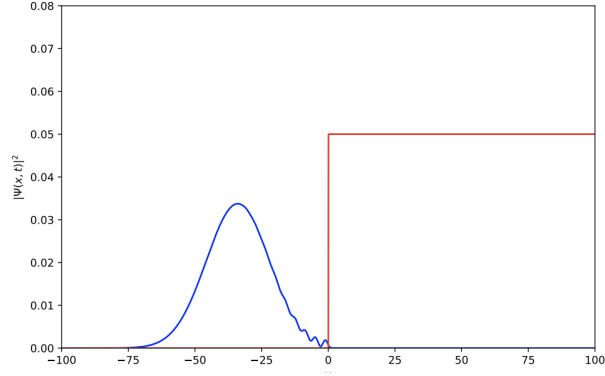


Figure 5:

We applied an electric field along the x-direction and introduced a new potential, $V(x)$, defined as follows: $V(x) = 0$ for $x < 0$ inside the metal, $V(x) = V_0 (1 - \frac{x}{L})$ for $0 < x < L$, and $V(x) = 0$ for $x > L$ outside the metal, where $L = 4$. We plotted $\ln T$ against $V_0^{3/2}$ for $V_0 = 0.6, 0.7, 0.8$. Figure 6 shows a linear relationship, which agrees with theory, as the transmission probability in a tunneling decreases exponentially with increasing potential barrier height.

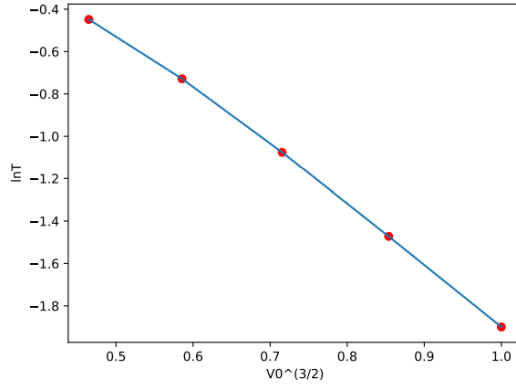


Figure 6: V_0